

Data Collection and Preprocessing Phase

Date	20 Sep 2026
Team ID	Master of Computer Applications
Project Title	Online Learning Engagement and Student Performance Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	Dataset Scope: The <code>online_education_dataset.csv</code> contains 32,593 student records across 14 attributes capturing demographic information (gender, region, highest_education, imd_band), academic load (studied_credits), platform activity metrics (total_clicks), academic outcome metrics (avg_score, final_result, pass_flag, dropout_flag), and behavioral indicators (engagement_level, performance_level, risk_level).
Data Cleaning	Missing Values Handled: • Imputed <code>imd_band</code> with "Unknown". • Filled missing <code>total_clicks</code> with 0 and re-evaluated <code>engagement_level</code> as "None". • Filled missing <code>avg_score</code> with student median score

	(or overall mean for unassessed students) and updated missing performance_level to "Unassessed". Duplicates: Removed 3,808 duplicate id_student records to retain unique student instances.
Data Transformation	Power Query Operations Applied: • Filtering: Excluded test records where studied_credits was less than 30 credits. • Pivoting & Dynamic Aggregation: Pivoted assessment scores to calculate overall course averages. • Calculated Columns: Added click_per_credit ratio (<i>total_clicks/studied_credits</i>) and mapped custom risk scoring metrics
Data Type Conversion	Schema Rectification: • id_student: Converted to Text/String format to avoid integer aggregation. • pass flag & dropout_flag: Cast to Boolean flags (1/0). • total_clicks & avg_score: Explicitly cast to Decimal/Float types. • imd_band, gender, region: Categorical text fields.
Column Splitting and Merging	Structural Refinement: • Splitting: Split imd_band (e.g., 20-30%) into imd_min_pct (20%) and imd_max_pct (30%) numerical boundary columns. • Merging: Combined gender and highest_education into a consolidated demographic key Demographic_Segment (e.g., M_HE Qualification).
Data Modeling	Relationships & Measures: • Built a Star Schema model linking a central

	Fact_StudentPerformance table to dimension tables: Dim_Student, Dim_Demographics, and Dim_Region. • Set up 1:N single-direction relationships from Dimensions to Fact. • Created DAX measures for aggregated retention, pass rates, and dynamic click ratios.
Save Processed Data	Storage & Future Use: Applied all Power Query M-steps, loaded the cleaned table structure into Power BI Desktop / SQL Database, and exported the finalized cleaned dataset as cleaned_online_education.csv for analytics.